

# Electricity Theft Detection for Smart Homes with Knowledge-Based Synthetic Attack Data

Abraham Olufemi Abiodun. Email: [abraham.olufemi.abiodun aj9@naist.ac.jp](mailto:abraham.olufemi.abiodun aj9@naist.ac.jp)



## Abstract

Electricity thefts are conventionally manually detected by inspections, accusations, and the failure of meters. However, the recent evolution of machine learning may allow the automatic detection of electricity theft only from the patterns of meter readings. Electric consumption heavily relies on many factors, e.g., the lifestyle of the day and the weather, and thus the accuracy of detection is questioned. We propose an electricity theft detection framework for smart homes with knowledge-based synthetic attack data. This allows training of the attack classifier only from the legitimate power consumption data, i.e, without attack actions and associated labels. We identified five attack patterns as the knowledge which consisted of smart attacks and legacy attacks. We have conducted comprehensive evaluations with nine machine learning models using the Almanac of Minutely Power dataset version 2 (AMPds2) dataset fine-grained time-series data of a smart home. We found that Gradient Boosting-based algorithms achieved the best, and Random Forest performed alternatively with almost 100% accuracy for detecting and classifying legacy attacks. Some smart attacks were not detected, but those algorithms achieved good performance in detection and classification.

## Background & Motivation

- Electricity Theft Attack (ETA) common all over the world<sup>1</sup>
  - Developing countries mostly affected<sup>2</sup>
- Indian loses more than 1/5<sup>th</sup> of total electricity production to theft<sup>3</sup>
- ETA detected by regular inspection of power meters<sup>4</sup>

- Method too dependent on human and still have difficulties in finding electricity thefts because of complexity of power systems

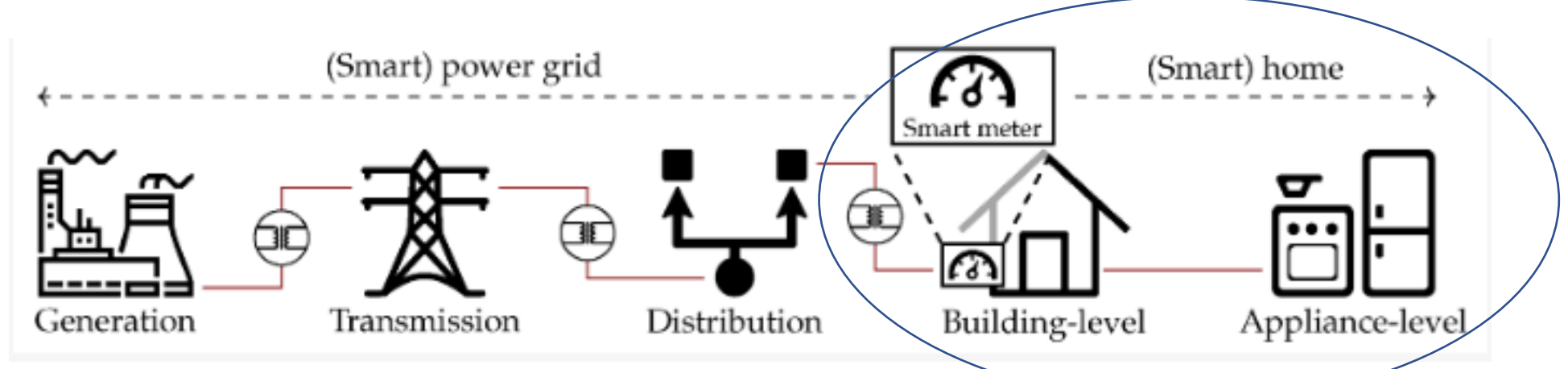
- Data manipulation

- FAI<sup>1</sup> in SM<sup>2</sup> home network

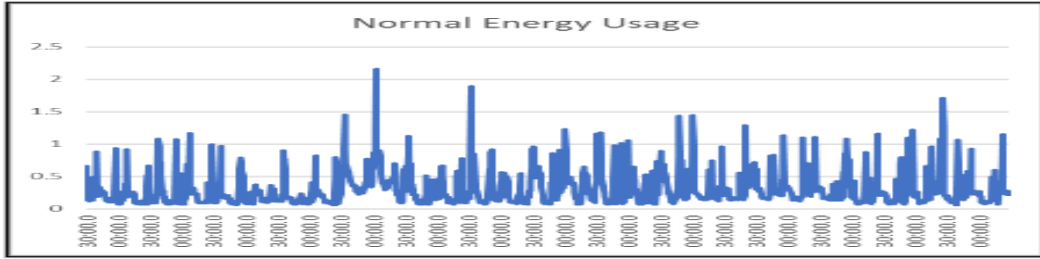
- Securing transactions (AMI)<sup>3</sup>

- Energy theft detection at the appliance level

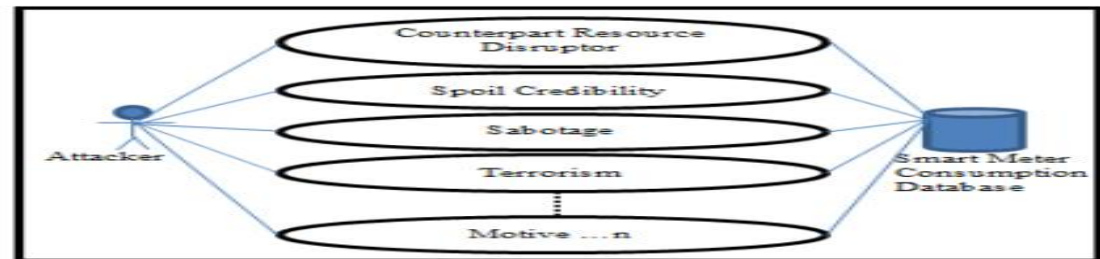
<https://www.spglobal.com/marketintelligence/en/news-insights/trending/abhemb-9pog6xryj18tia>, <https://www.prnewswire.com/news-releases/96-billion-is-lost-every-year-to-electricity-theft-300453411.html>  
<https://epower.com/news/world-loses-89.3-billion-annually-to-electricity-theft/#>



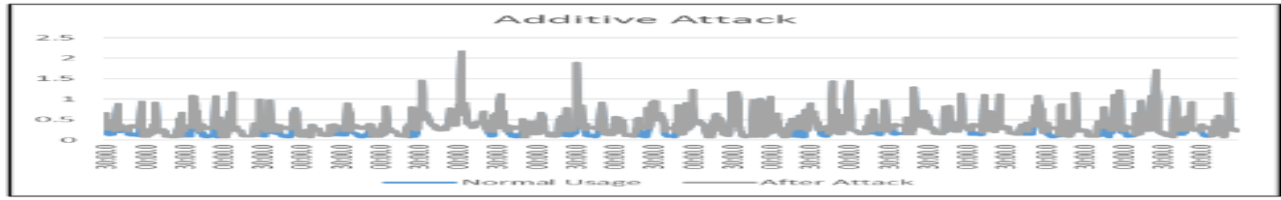
### TAXONOMY OF ADDITIVE AND DEDUCTIVE ATTACK ON SMART METER DATA



Readings of normal usage of Smart Meter



Additive attack motivation



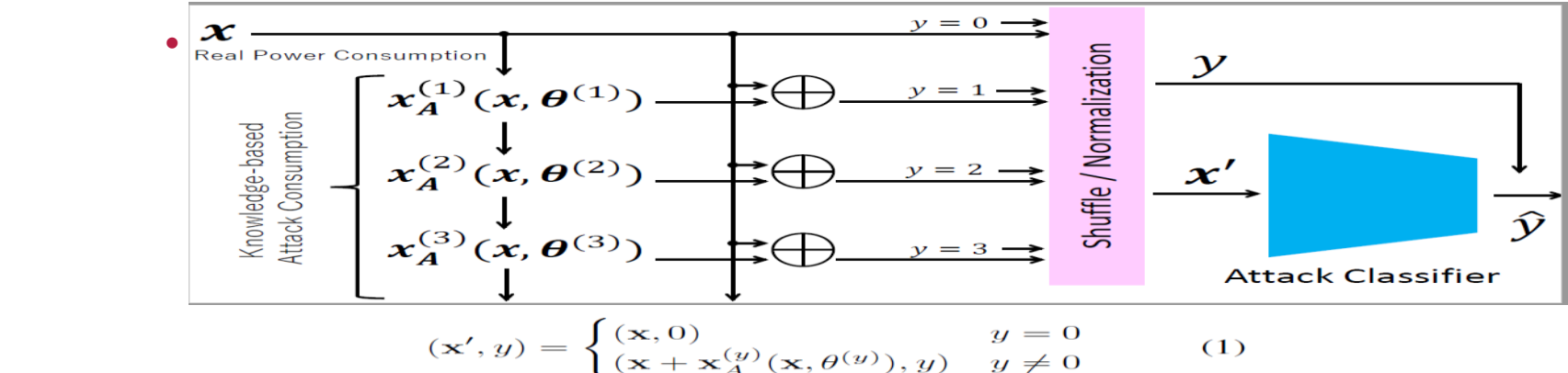
Readings of normal usage after additive attack

## Proposed Idea and Methodology

### ATTACK CLASSIFICATION

- Attacker may physically access the power distribution board
- Steal power to make the household pay the bills
- Attacks identified and classified as: Smart attacks & Legacy attacks:
  - Baseload:Attack: Baseload theft of 100W for 24hrs
  - Weakload Attack: Weakload theft of 500W at (threshold 100W)
  - Peakhour Attack: Peakhour theft of 2000W at (threshold 500W)
  - Midnight attack: Midnight theft of 1000W randomly at 1:00am – 3:00am
  - Evil-Twin Attack: Evil-Twin theft of aggregated power consumption of two houses

- Synthetic attack data to validate Machine Learning (ML)



Collection of (x',y) can be applied to Supervised ML model

## DATA SOURCE

Table I: Names of the appliances monitoring points

| Point ID | Description           | Point ID | Description              |
|----------|-----------------------|----------|--------------------------|
| WIE      | Whole House Meter     | GRE      | Garage/Furnace           |
| B1E      | North Bedroom         | HPE      | Heat Pump                |
| B2E      | Master/South BR       | HTE      | Instant Hot Water Unit   |
| BME      | Basement              | OFE      | Home Office              |
| CDE      | Plug/Lights           | OUE      | Outside Plug             |
| CWE      | Clothes Dryer         | TVE      | Entertainment TV/PVR/AMP |
| DNE      | Dinning Room Plugs    | UTE      | Utility Room             |
| DWE      | Dishwasher            | WOE      | Plug Wall Oven           |
| EBE      | Electronics Workbench | UNE      | Unmetered Loads          |
| EQE      | Security/Network      |          |                          |
| EGE      | Kitchen Fridge        |          |                          |

- <sup>1</sup>AMPds2 (Almanac of Minutely Power Dataset version2)

<sup>1</sup>S. Makonin, B. Ellert, I. V. Bajic, and F. Popowich, "Electricity, water, and natural gas consumption of a residential house in canada from 2012 to 2014," Scientific data, vol. 3, no. 1, pp. 1–12, 2016.

## RESULTS

### EVALUATION

TABLE V  
EXPERIMENTAL RESULTS OF DIFFERENT ALGORITHMS. THE RANKING IS BASED ON THE ACCURACY OF HOME A.

| Ranking | Models                          | Home | Accuracy | Precision | Recall | F1-score |
|---------|---------------------------------|------|----------|-----------|--------|----------|
| 1       | Gradient Boosting(GB)           | A    | 92.98%   | 0.94      | 0.93   | 0.93     |
|         |                                 | B    | 94.53%   | 0.95      | 0.95   | 0.95     |
|         |                                 | C    | 93.61%   | 0.94      | 0.94   | 0.93     |
| 2       | Extreme Gradient Boosting(XGB)  | A    | 92.53%   | 0.94      | 0.93   | 0.92     |
|         |                                 | B    | 95.00%   | 0.96      | 0.95   | 0.95     |
|         |                                 | C    | 92.75%   | 0.94      | 0.93   | 0.92     |
| 3       | Random Forest (RF)              | A    | 86.78%   | 0.91      | 0.87   | 0.87     |
|         |                                 | B    | 89.19%   | 0.92      | 0.89   | 0.90     |
|         |                                 | C    | 80.47%   | 0.82      | 0.80   | 0.80     |
| 4       | Decision Tree Classifier(DTC)   | A    | 70.58%   | 0.75      | 0.71   | 0.71     |
|         |                                 | B    | 77.33%   | 0.81      | 0.77   | 0.78     |
|         |                                 | C    | 77.52%   | 0.78      | 0.78   | 0.78     |
| 5       | Multilayer Perceptron 4 (MLP-4) | A    | 66.44%   | 0.70      | 0.66   | 0.67     |
|         |                                 | B    | 71.28%   | 0.76      | 0.71   | 0.72     |
| 6       | Multilayer Perceptron 3 (MLP-3) | A    | 65.75%   | 0.69      | 0.66   | 0.66     |
|         |                                 | B    | 69.88%   | 0.73      | 0.70   | 0.71     |
|         |                                 | C    | 71.01%   | 0.76      | 0.71   | 0.72     |
| 7       | Logistic Regression (LR)        | A    | 54.83%   | 0.54      | 0.55   | 0.54     |
|         |                                 | B    | 56.86%   | 0.56      | 0.57   | 0.55     |
|         |                                 | C    | 70.76%   | 0.73      | 0.71   | 0.71     |
| 8       | Ridge Regression (RRC)          | A    | 52.30%   | 0.53      | 0.52   | 0.52     |
|         |                                 | B    | 54.88%   | 0.56      | 0.55   | 0.54     |
|         |                                 | C    | 62.04%   | 0.67      | 0.62   | 0.62     |
| 9       | Support Vector Machine(SVM)     | A    | 52.20%   | 0.51      | 0.52   | 0.51     |
|         |                                 | B    | 56.16%   | 0.56      | 0.56   | 0.55     |
|         |                                 | C    | 67.69%   | 0.69      | 0.68   | 0.67     |

Ranking of algorithms based on the accuracy of Home A.

| Color | Description                          | Rank                              |
|-------|--------------------------------------|-----------------------------------|
| Red   | Overall accuracy more than 80%       | (1) GB<br>(2) XGB<br>(3) RF       |
| Green | Overall accuracy between 70% - 80%   | (4) DTC<br>(5) MPL_4<br>(6) MPL_3 |
| Blue  | Overall accuracy is between 60% -70% | (7) LR<br>(8) RRC<br>(9) SVM      |
| Gray  | Overall accuracy is less than 60%    |                                   |

- Midnight and Evil-Twin attacks were well classified in all the classifiers.
- GB, XGB, and RF have achieved 100% accuracy for these attacks.

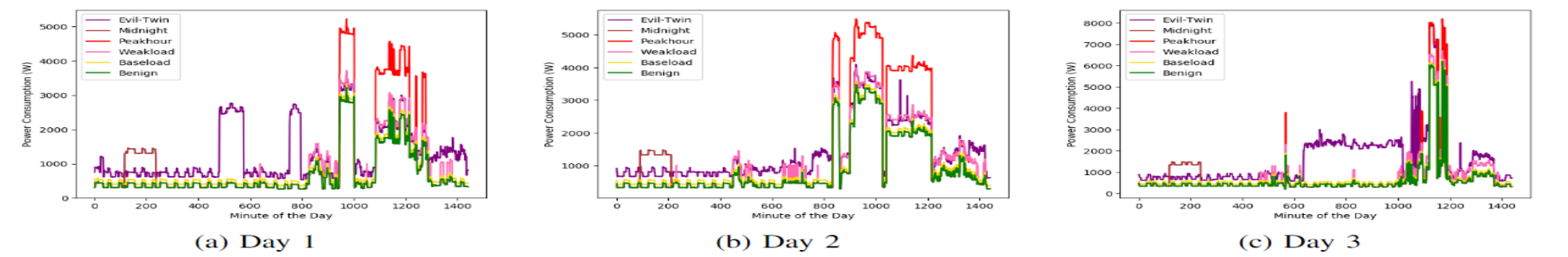


Fig. 3. Electricity usage of Home A with synthetic attack data on different days. The benign consumption patterns drastically change day by day. Especially, the Weakload and Peakhour attacks are sophisticated – having similar consumption patterns with the legitimate usage although the values are different.

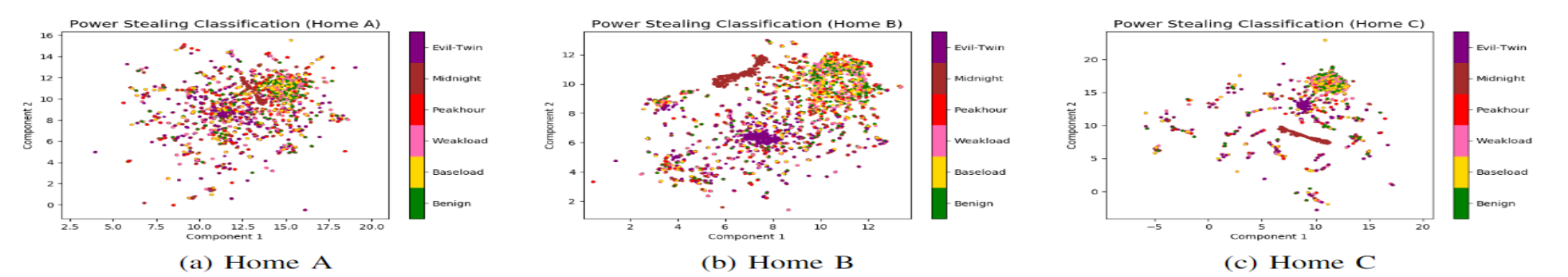


Fig. 4. 2D projections of attack contained electricity usage with the benign case by uniform manifold approximation and projection (UMAP) [18]. Each plot corresponds to the consumption pattern of a day on different attack scenarios. Depending on the available appliances of the homes, the shapes of the attack classes are very different. There are more overlaps among classes in Home A, whereas we observe fewer overlaps in Home C.

## Conclusion and Future Work

- Developed a detection framework for identifying electricity theft in smart homes, using five defined synthetic attack types.
- Achieved high detection accuracy with Gradient Boosting and Random Forest algorithms, notably for legacy attacks.
- Identified the need for improved detection of sophisticated 'smart' attack methods.
- Future research to expand attack models, analyze varied household data, and explore the correlation between attack size and detection success